Accelerated MD TorchSim timing scaling with system size

Points: systems; dashed: fits (n=255; 7 invalid skipped)

NequIP-OAM-L n=13

orb-v3-direct-20-omat n=14

MACE-MP-0 n=13

orb-v2 n=14

GRACE-2L-OAM n=14

MACE-MPA-0 n=13

MatterSim-v1.0.0-5M n=14

MACE-MH-1-OMAT n=13

GRACE-2L-MPtrj n=13

orb-v3-conservative-inf-omat n=14

UMA-S-P1 (turbo) n=14

UMA-S-P1 (compile) n=14

CHGNet n=13

EquiformerV2 n=13

PET-OAM-XL n=14

PET-OMAT-XL n=14

UMA-M-P1 (turbo) n=13

UMA-M-P1 (compile) n=13

eSEN-30M-OAM n=12

MD time per step [ms] (log scale)

System size [number of atoms] (log scale)